CHALLENGES

IT systems have become unmanageably complex and increasingly costly to maintain.

IT systems are hindering the organization's ability to respond to current, and future, market conditions in a timely and cost-effective manner.

Mission-critical information is consistently out-of-date and/or just plain wrong.

A culture of distrust between the business and technology sides of the organization.

The ReesanGroup

Legacy systems challenges in digital transformation: Reesan Group

Overview

ReesanGroup, a chain of large retail stores has grown by acquisition over the past few years. Its core competence has been in its ability to keep operating costs low and pass these savings onto customers. Its operations were supported by fit for purpose ERP, Jupiter.

However, with this growth came an architectural misalignment as integration of newly acquired businesses added complexity to its operating environment. Further, rationalism of business functions and staffs created complications to the management and operational structures.

Because of these issues ReesanGroup's ability to operate in a very competitive and highly regulated marketplace is challenged.

Further issues are faced associated with the need to embrace 'disruptive' technologies being hampered by legacy systems and work practices and ReesanGroup lagging behind its competitors and loosing competitive ground.

Introduction

ReesanGroup is a chain of large retail stores. It started as a regional chain in 1990. In 2005, it developed an innovative software system that enabled it to run retail stores very efficiently. It called this system Jupiter. Jupiter incorporated some innovate business ideas, such as customer-relationship management, inventory management, automated retail billing, and even utility optimization.

It consisted of 50 stores distributed in five major populations areas each served by a centralised warehouse,

Jupiter consisted of three major applications: Jupiter/Store, which ran on small computers at retail stores; Jupiter/Warehouse, which ran on servers in regional warehouses; and Jupiter/Head, which ran on a large server at the head office.

These three programs communicated through files that were transferred from one location (for example, a store) to another (for example, a regional warehouse). When reliable communications lines existed, file transfers could occur through FTP. The system was also flexible enough to accommodate transfers through secure protocol (SFTP), where necessary.

By 2012, ReesanGroup was doing quite well—in part, because of the cost-cutting moves enabled by the Jupiter system. ReesanGroup decided to begin expansion. To do this, it purchased six regional retail chains. With these purchases, ReesanGroup extended its reach to all over Australia.

WHY ENTERPRISE ARCHITECTURE?

Improve the alignment of information systems and technologies with the mission/business needs of the organization.

Provide a common language to facilitate linkages and communication between complex architectures and applications.

Provide a unifying approach to minimize the effort of data collection, storage and access.

Facilitate efficient identification of necessary changes and change implementation – identifies reuse opportunities.

Organisational Impact

ReesanGroup's acquisitions and the new staff and structures that were brought on board became decreasingly effective due to:

- Senior managers from both existing and newly acquired organisations became politically motivated in order to maintain powerbases as integration proceeded.
- There was tension between staff groups with each group clinging on to legacy practices resulting in sub optimum re-engineering outcomes and an ineffective deployment of staff resources.
- Managers in senior roles in the Shared Services Functions i.e. Finance, Procurement, Human Resource Management, Information Systems (ICT) became protective of their legacy systems and work practices which created duplication and functional overlaps.
- The extent of change and the inevitable redundancies those arose had a severe impact on staff moral which negatively impacted productivity and the ability to respond to change.

Architectural Impact

By 2014, the ReesanGroup executives started facing difficulties in managing their business operations. They were realizing that the same software systems that had initially fueled ReesanGroup's success were now hampering its future. Some of the problems ReesanGroup was running into were the following:

- Jupiter/Store required regional specializations. For example, different sales plans needed to be supported in different regions due to climate and demographic variations, and these all required changes to the Jupiter/Store module.
- The regional warehouses that had been acquired through acquisition each had different ways of receiving orders from the retail stores and different procedures from ordering supplies from the wholesalers. Each of these differences required changes to the ReesanGroup/Warehouse module.
- The file-transfer approach to information sharing that had worked so well when ReesanGroup consisted of 50 retail stores, 5 regional warehouses, and one head office were turning out to be difficult to coordinate among 500 retail stores, 50 regional warehouses, 6 geographic offices, and one head office. This was due to the lack of standardized approach for file-transfer across all business applications. Consequently, files were often delivered late, sometimes not at all, and occasionally multiple times. This made it difficult for the head office to access reliable, up-to-date financial information, especially in the areas of sales and inventory.

It was clear to ReesanGroup management that the Jupiter system needed many enhancements. However, upgrading this system was difficult. Each of the three modules (store, warehouse, and head office) was huge, inefficient, and cumbersome, and each included functionality for everything that each entity might need.

The modules had grown to over 1 million lines of code each. It was difficult to change one function without affecting others. All of the functions accessed a single database, and changes to one record definition could ripple through the system in an unpredictable fashion. Changing even a single line of code had a ripple effect and required a rebuild of the entire multimillion-line module.

Jupiter had become almost impossible to manage. Debugging was difficult. Software builds were torturous. Installing new systems was hugely disruptive. And there were integration issues impacting on the integrity, performance and availability of data needed to make management decisions.

These technical problems soon created internal conflicts within the head office of ReesanGroup. The business executives of ReesanGroup wanted to acquire two more regional chains, but IT was still struggling to bring the existing acquisitions online.

This resulted in a rapidly growing divide between the business and the technical sides of ReesanGroup. The business side saw IT as reducing business agility. The technical side saw the business side as making impossible demands and blamed it for refusing to consult IT before entering into acquisition discussions.

The distrust had reached such a point that, by 2018, the CIO was no longer considered part of the executive team of ReesanGroup. The business side distrusted IT and tried to circumvent it at every opportunity. The technical side built its IT systems with little input from the business folks. Several large and expensive IT initiatives were ignored by the business side and were eventually abandoned.

The Tipping Point

External consultants and audits were increasingly, identifying operational fragmentation and loss of integrity in process and information as major factors in ReesanGroup's malaise. Although not externally validated ReesanGroup' organsiastional maturity had slipped from a managed level (level 4) of capability to displaying features of initial immature (level 1) capability. ReesanGroup seemed to be powerless to act.

By 2020, the inevitable happened and ReesanGroup was in crisis. It clearly needed to revamp its technical systems to make them easier to specialize for regional requirements. This was going to be an expensive proposition, and ReesanGroup couldn't afford for the effort to fail.

Just as importantly, ReesanGroup also had to rebuild its internal relationships. The constant bickering and distrust between business and IT was affecting morale, efficiency, and profitability of the organization. A company that only five years earlier was an industry leader in profitability—in large part, because of its innovative use of IT—was now struggling to stay out of the red—in large part, because of the inflexibility of IT systems.

ARCHITECTURAL ARTIFACT

A specific document, report, analysis, model, or other tangible that contributes to an architectural description.

ARCHITECTURAL TAXONOMY

A methodology for organizing and categorizing architectural artifacts.

Time to Act

Victoria, the CEO of Reesan Group, desperately needed a solution. At a CEO conference, she heard how many of her peers were using contemporary digital platforms to build stronger partnerships between their technical and business groups and deliver more cost-effective IT systems that enabled business agility and provided the ability to exploit disruptive technologies: Cloud Computing and Big Data in particular.

Victoria decided that this approach merited further investigation. She asked Steven, her CIO, to prepare a recommendation on the use of an integrated technical plaform within ReesanGroup. Steven was impressed with the approach, but recognized that any such initiative needed to be driven from the top and needed to involve the business side from the start. On Steven's recommendation, Victoria called a meeting with Smith, the Vice-President of Business, and Steven. Victoria announced that she had decided to create a common enterprise architecture for ReesanGroup that would unite its technical and business people. This common enterprise architecture would be named ReesanGroup-Enterprise Architecture. After it was completed, this platform would drive all new IT investment and ensure that every dollar invested in IT was delivering the maximum value to the business.

Victoria knew that was a bet-the-company decision for ReesanGroup. The digital transformation vision had to work. Victoria was depending on Smith (the business side) and Steven (the IT side) to make it work.

So, that is a complex business problem that requires as a solution. You (as an IS Consultant) will apply consulting concepts to provide a solution for ReesanGroup considering all aspects of business, technology and external market forces.

The Case Study

This case study has been prepared by Dr. Rod Dilnutt and Dr. Humza Naseer for teaching purposes and synthesises issues commonly found in business and industry.

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